

EXPERIMENT 4

AIM:

To count the frequency of occurrence of words in a given body of text using Python.

CODE:

```
import nltk

from nltk.tokenize import word_tokenize
from nltk.corpus import Gutenberg

nltk.download('gutenberg')
nltk.download('punkt')

sample = gutenberg.raw("carroll-alice.txt")

token = word_tokenize(sample)

wlist = []

for i in range(50):
    wlist.append(token[i])

wordfreq = [wlist.count(w) for w in wlist]

# Print the word-frequency pairs
print("Pairs\n" + str(list(zip(wlist, wordfreq))))
```

OUTPUT:

Pairs

```
[(['', 1), ('Alice', 2), ('"s"', 1), ('Adventures', 1), ('in', 1), ('Wonderland', 1), ('by', 2), ('Lewis', 1), ('Carroll', 1), ('1865', 1), (']', 1), ('CHAPTER', 1), ('I', 1), ('.', 1), ('Down', 1), ('the', 3), ('Rabbit-Hole', 1), ('Alice', 2), ('was', 1), ('beginning', 1), ('to', 2), ('get', 1), ('very', 1), ('tired', 1), ('of', 2), ('sitting', 1), ('by', 2), ('her', 2), ('sister', 1), ('on', 1), ('the', 3), ('bank', 1), ('.', 1), ('and', 1), ('of', 2), ('having', 1), ('nothing', 1), ('to', 2), ('do', 1), (':', 1), ('once', 1), ('or', 1), ('twice', 1), ('she', 1), ('had', 1), ('peeped', 1), ('into', 1), ('the', 3), ('book', 1), ('her', 2)]
```

RESULT:

The frequency of occurrence of words in the given text was successfully calculated using Python and displayed as word-frequency pairs.